

EDEXCEL INTERNATIONAL A LEVEL

# WMA11/01 October 2022 Worked Solutions

Oct 2022 paper rebuilt with the worked solution after each question.

**9**

paper questions

**WMA11**

Pure 1

**Single-  
paper  
solutions**standalone IAL  
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - [eliteigcse.com](https://eliteigcse.com)

**P1**

**PAST PAPER**

# **WMA11/01 October 2022**

**Oct 2022 | 9 questions | 75 marks**

**9**

questions

**75**

marks

**Answers**

worked solution  
after each  
question

## Question 1

## Differentiation

1. The curve  $C$  has equation

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} \quad x > 0$$

- (a) Find  $\frac{dy}{dx}$ , giving your answer in simplest form.

(3)

The point  $R\left(2, \frac{13}{2}\right)$  lies on  $C$ .

- (b) Find the equation of the tangent to  $C$  at the point  $R$ . Write your answer in the form  $ax + by + c = 0$ , where  $a$ ,  $b$  and  $c$  are integers to be found.

(3)

(Total for Question 1 is 6 marks)

## Worked Solution - Question 1

**1. Rewrite the curve**

$$y = \frac{x^3}{4} - x^2 + \frac{17}{x} = \frac{1}{4}x^3 - x^2 + 17x^{-1}.$$

**2. Differentiate**

$$\frac{dy}{dx} = \frac{3}{4}x^2 - 2x - 17x^{-2} = \frac{3}{4}x^2 - 2x - \frac{17}{x^2}.$$

**3. Find the gradient at R**

At  $x = 2$ , the tangent gradient is  $\frac{3}{4}(4) - 2(2) - \frac{17}{4} = 3 - 4 - \frac{17}{4} = -\frac{21}{4}.$

**4. Use the point R**

The point is  $R\left(2, \frac{13}{2}\right)$ , so  $y - \frac{13}{2} = -\frac{21}{4}(x - 2).$

**5. Write in integer form**

Multiplying by 4 gives  $4y - 26 = -21x + 42$ , hence  $21x + 4y - 68 = 0.$

**Final answer**

(a)  $\frac{dy}{dx} = \frac{3}{4}x^2 - 2x - \frac{17}{x^2}.$

(b)  $21x + 4y - 68 = 0.$

## Question 2

## Integration

2. Given that

$$(x - 5)(2x + 1)(x + 3) \equiv ax^3 + bx^2 - 32x - 15$$

where  $a$  and  $b$  are constants,

(a) find the value of  $a$  and the value of  $b$ .

(2)

(b) Hence find

$$\int \frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} \, dx$$

writing each term in simplest form.

(5)

(Total for Question 2 is 7 marks)

## Worked Solution - Question 2

**1. Expand the identity**

$$(2x + 1)(x + 3) = 2x^2 + 7x + 3, \text{ so}$$

$$(x - 5)(2x^2 + 7x + 3) = 2x^3 - 3x^2 - 32x - 15.$$

**2. Read off a and b**

Comparing with  $ax^3 + bx^2 - 32x - 15$  gives  $a = 2$  and  $b = -3$ .

**3. Rewrite the integrand**

$$\frac{(x - 5)(2x + 1)(x + 3)}{5\sqrt{x}} = \frac{1}{5}(2x^{5/2} - 3x^{3/2} - 32x^{1/2} - 15x^{-1/2}).$$

**4. Integrate term by term**

$$\int \frac{1}{5}(2x^{5/2})dx = \frac{4}{35}x^{7/2} \text{ and } \int \frac{1}{5}(-3x^{3/2})dx = -\frac{6}{25}x^{5/2}.$$

**5. Complete the integral**

$$\int \frac{1}{5}(-32x^{1/2})dx = -\frac{64}{15}x^{3/2} \text{ and } \int \frac{1}{5}(-15x^{-1/2})dx = -6x^{1/2}, \text{ then add } c.$$

**Final answer**

$$(a) a = 2, b = -3.$$

$$(b) \frac{4}{35}x^{7/2} - \frac{6}{25}x^{5/2} - \frac{64}{15}x^{3/2} - 6\sqrt{x} + c.$$

## Question 3

Straight Line

3. The share price of a company is monitored.

Exactly 3 years after monitoring began, the share price was £1.05

Exactly 5 years after monitoring began, the share price was £1.65

The share price, £ $V$ , of the company is modelled by the equation

$$V = pt + q$$

where  $t$  is the number of years after monitoring began and  $p$  and  $q$  are constants.

- (a) Find the value of  $p$  and the value of  $q$ .

(3)

Exactly  $T$  years after monitoring began, the share price was £2.50

- (b) Find the value of  $T$ , according to the model, giving your answer to one decimal place.

(2)

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(Total for Question 3 is 5 marks)

## Worked Solution - Question 3

**1. Use the two data points**

The model  $V = pt + q$  passes through  $(3, 1.05)$  and  $(5, 1.65)$ .

**2. Find the gradient**

$$p = \frac{1.65 - 1.05}{5 - 3} = \frac{0.60}{2} = 0.30.$$

**3. Find the intercept**

Using  $1.05 = 0.30(3) + q$ , we get  $q = 0.15$ .

**4. Find T**

For  $V = 2.50$ ,  $2.50 = 0.30T + 0.15$ , so  $T = \frac{2.35}{0.30} = 7.833 \dots$

**5. Round correctly**

To one decimal place,  $T = 7.8$ .

**Final answer**

(a)  $p = 0.3$ ,  $q = 0.15$ .

(b)  $T = 7.8$  years.



**Question 4****Indices & Surds****4.**

**In this question you must show detailed reasoning.  
Solutions relying on calculator technology are not acceptable.**

$$f(x) = x^2(2x + 1) - 15x$$

(a) Solve

$$f(x) = 0 \quad (4)$$

(b) Hence solve

$$y^{\frac{4}{3}}\left(2y^{\frac{2}{3}} + 1\right) - 15y^{\frac{2}{3}} = 0 \quad y > 0$$

giving your answer in simplified surd form. (2)

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**(Total for Question 4 is 6 marks)**

## Worked Solution - Question 4

**1. Factor  $f(x)$** 

$$f(x) = x^2(2x + 1) - 15x = x(2x^2 + x - 15).$$

**2. Factor the quadratic**

$$2x^2 + x - 15 = (2x - 5)(x + 3).$$

**3. Solve  $f(x) = 0$** 

$$x(2x - 5)(x + 3) = 0, \text{ so } x = 0, x = \frac{5}{2}, \text{ or } x = -3.$$

**4. Use the substitution for part (b)**

The equation is the same as  $f(u) = 0$  with  $u = y^{2/3}$ .

**5. Choose the valid root**

Since  $y > 0$ ,  $u = y^{2/3} > 0$ . The valid root is  $u = \frac{5}{2}$ , so  $y^{2/3} = \frac{5}{2}$ .

**6. Write  $y$  in surd form**

$$y = \left(\frac{5}{2}\right)^{3/2} = \sqrt{\frac{125}{8}} = \frac{5\sqrt{10}}{4}.$$

**Final answer**

$$(a) x = -3, 0, \frac{5}{2}.$$

$$(b) y = \frac{5\sqrt{10}}{4}.$$

## Question 5

## Integration

5.

**In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.**

The curve  $C$  has equation  $y = f(x)$ ,  $x > 0$

Given that

- $f'(x) = \frac{12}{\sqrt{x}} + \frac{x}{3} - 4$
- the point  $P(9, 8)$  lies on  $C$

(a) find, in simplest form,  $f(x)$

(5)

The line  $l$  is the normal to  $C$  at  $P$

(b) Find the coordinates of the point at which  $l$  crosses the  $y$ -axis.

(4)

**(Total for Question 5 is 9 marks)**

## Worked Solution - Question 5

**1. Integrate  $f'(x)$** 

$$f'(x) = 12x^{-1/2} + \frac{x}{3} - 4, \text{ so } f(x) = 24\sqrt{x} + \frac{x^2}{6} - 4x + c.$$

**2. Use  $P(9, 8)$** 

$$8 = 24\sqrt{9} + \frac{9^2}{6} - 4(9) + c = 72 + \frac{81}{6} - 36 + c.$$

**3. Find  $c$** 

$$72 + \frac{81}{6} - 36 = \frac{99}{2}, \text{ so } c = 8 - \frac{99}{2} = -\frac{83}{2}.$$

**4. Find the tangent gradient at  $P$** 

$$f'(9) = \frac{12}{3} + \frac{9}{3} - 4 = 3.$$

**5. Find the normal gradient**

The normal gradient is the negative reciprocal of 3, so it is  $-\frac{1}{3}$ .

**6. Find the y-intercept**

The normal through  $(9, 8)$  is  $y - 8 = -\frac{1}{3}(x - 9)$ , so  $y = -\frac{1}{3}x + 11$ . It crosses the y-axis at  $(0, 11)$ .

**Final answer**

$$(a) f(x) = 24\sqrt{x} + \frac{x^2}{6} - 4x - \frac{83}{2}.$$

$$(b) (0, 11).$$

## Question 6

## Graphs of Functions

6. (a) Given that  $k$  is a positive constant such that  $0 < k < 4$  sketch, on **separate axes**, the graphs of

(i)  $y = (2x - k)(x + 4)^2$

(ii)  $y = \frac{k}{x^2}$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of  $k$ , where appropriate.

(5)

- (b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)

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(Total for Question 6 is 6 marks)

## Worked Solution - Question 6

Topic group

**1. Key points for  $y = (2x-k)(x+4)^2$** 

The graph has x-intercepts  $(-4, 0)$  and  $\left(\frac{k}{2}, 0\right)$ . The root at  $x = -4$  is a repeated root, so the graph touches the x-axis there.

**2. Use  $0 < k < 4$** 

Since  $0 < k < 4$

**3. Key features for  $y = k/x^2$** 

The graph is positive on both sides of the y-axis, with asymptotes  $x = 0$  and  $y = 0$ . It has no intercepts with the coordinate axes.

**4. Compare the two graphs**

The equation  $(2x - k)(x + 4)^2 = \frac{k}{x^2}$  represents intersections between the two graphs.

**5. State the number of roots**

From the sketches, the positive branch of  $\frac{k}{x^2}$  meets the increasing positive branch of  $(2x - k)(x + 4)^2$  once only. Therefore there is one root.

**Final answer**

The graphs intersect once, so the equation has one root.

## Question 7

## Transformations

7.

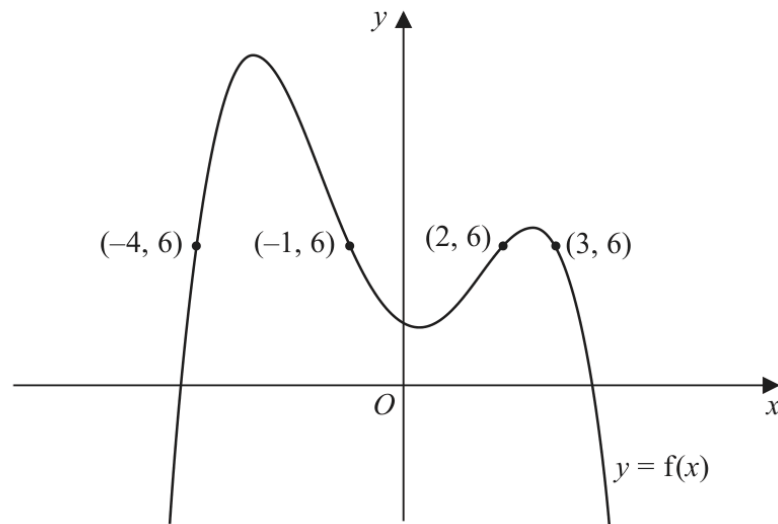


Figure 1

Figure 1 shows the curve with equation  $y = f(x)$ .

The points  $P(-4, 6)$ ,  $Q(-1, 6)$ ,  $R(2, 6)$  and  $S(3, 6)$  lie on the curve.

(a) Using Figure 1, find the range of values of  $x$  for which

$$f(x) < 6 \quad (3)$$

(b) State the largest solution of the equation

$$f(2x) = 6 \quad (1)$$

(c) (i) Sketch the curve with equation  $y = f(-x)$ .

On your sketch, state the coordinates of the points to which  $P$ ,  $Q$ ,  $R$  and  $S$  are transformed.

(ii) Hence find the set of values of  $x$  for which

$$f(-x) \geq 6 \text{ and } x < 0 \quad (4)$$

(Total for Question 7 is 8 marks)

## Worked Solution - Question 7

**1. Read  $f(x) < 6$  from the graph**

The curve is below the line  $y = 6$  for  $x < -4$ , for  $-13 < x < -4$ .

**2. Solve  $f(2x) = 6$** 

Since  $f(u) = 6$  at  $u = -4, -1, 2, 3$ , we set  $2x = -4, -1, 2, 3$ . The solutions are  $x = -2, -\frac{1}{2}, 1, \frac{3}{2}$ .

**3. State the largest solution**

The largest solution is  $x = \frac{3}{2}$ .

**4. Reflect the points in the y-axis**

For  $y = f(-x)$ , each point  $(x, y)$  becomes  $(-x, y)$ .

**5. List the transformed points**

$P(-4, 6) \rightarrow (4, 6)$ ,  $Q(-1, 6) \rightarrow (1, 6)$ ,  $R(2, 6) \rightarrow (-2, 6)$ , and  $S(3, 6) \rightarrow (-3, 6)$ .

**6. Use the reflected graph**

For  $x < 0$ , the part of the reflected graph on or above  $y = 6$  is from  $x = -3$  to  $x = -2$ .

**Final answer**

(a)  $x < -4$ ,  $-13 < x < -4$ .

(b)  $\frac{3}{2}$ .

(c)(ii)  $-3 \leq x \leq -2$ .



## Question 8

Radians

8.

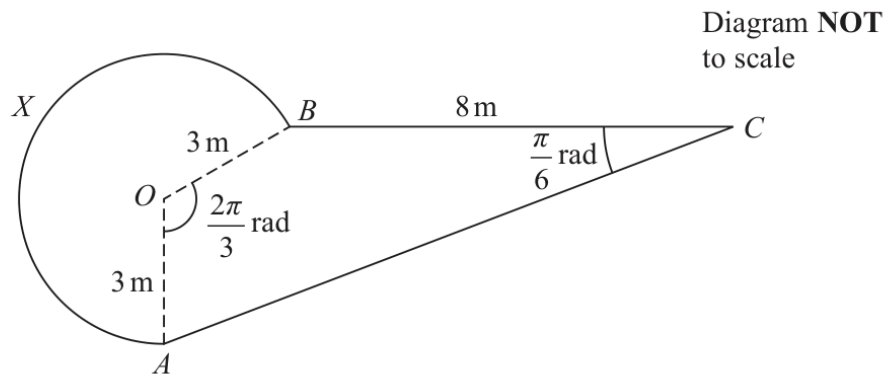


Figure 2

Figure 2 shows the plan view of a design for a pond.

The design consists of a sector  $AOBX$  of a circle centre  $O$  joined to a quadrilateral  $AOBC$ .

- $BC = 8$  m
- $OA = OB = 3$  m
- angle  $AOB$  is  $\frac{2\pi}{3}$  radians
- angle  $BCA$  is  $\frac{\pi}{6}$  radians

(a) Calculate (i) the exact area of the sector  $AOBX$ ,

(ii) the exact perimeter of the sector  $AOBX$ .

(5)

(b) Calculate the exact area of the triangle  $AOB$ .

(2)

(c) Show that the length  $AB$  is  $3\sqrt{3}$  m.

(2)

(d) Find the total surface area of the pond. Give your answer in  $\text{m}^2$  correct to 2 significant figures.

(5)

(Total for Question 8 is 14 marks)

## Worked Solution - Question 8

**1. Find the angle of the sector AOBX**

The sector AOBX is the major sector, so its angle is  $2\pi - \frac{2\pi}{3} = \frac{4\pi}{3}$ .

**2. Find the sector area**

$$\text{Area} = \frac{1}{2}r^2\theta = \frac{1}{2}(3^2)\left(\frac{4\pi}{3}\right) = 6\pi \text{ m}^2.$$

**3. Find the sector perimeter**

The arc length is  $r\theta = 3\left(\frac{4\pi}{3}\right) = 4\pi$ . Adding the two radii gives  $4\pi + 6 \text{ m}$ .

**4. Find the area of triangle AOB**

$$\text{Area} = \frac{1}{2}(3)(3)\sin\left(\frac{2\pi}{3}\right) = \frac{9}{2} \cdot \frac{\sqrt{3}}{2} = \frac{9\sqrt{3}}{4} \text{ m}^2.$$

**5. Show AB**

By the cosine rule,  $AB^2 = 3^2 + 3^2 - 2(3)(3)\cos\left(\frac{2\pi}{3}\right) = 18 + 9 = 27$ , so  $AB = 3\sqrt{3}$ .

**6. Find AC**

In triangle ABC,  $AB^2 = AC^2 + 8^2 - 2(AC)(8)\cos\left(\frac{\pi}{6}\right)$ , so  $27 = AC^2 - 8\sqrt{3}AC + 64$ .

**7. Choose the correct length**

Solving gives  $AC = 4\sqrt{3} \pm \sqrt{11}$ . From the diagram,  $AC$  is the longer value, so  $AC = 4\sqrt{3} + \sqrt{11}$ .

### 8. Find the total area

Area of triangle ABC is  $\frac{1}{2}(8)(AC) \sin\left(\frac{\pi}{6}\right) = 2AC = 8\sqrt{3} + 2\sqrt{11}$ . Total area is  $6\pi + \frac{9\sqrt{3}}{4} + 8\sqrt{3} + 2\sqrt{11} = 43.236\dots$ , so  $43 \text{ m}^2$  to 2 significant figures.

#### Final answer

(a)(i)  $6\pi \text{ m}^2$ , (ii)  $4\pi + 6 \text{ m}$ .

(b)  $\frac{9\sqrt{3}}{4} \text{ m}^2$ .

(c)  $AB = 3\sqrt{3} \text{ m}$ .

(d)  $43 \text{ m}^2$ .

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14 marks

## Question 9

Differentiation

9.

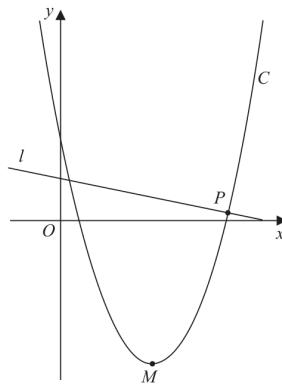


Figure 3

Figure 3 shows a sketch of the curve  $C$  with equation

$$y = \frac{1}{2}x^2 - 10x + 22$$

- (a) Write  $\frac{1}{2}x^2 - 10x + 22$  in the form

$$a(x + b)^2 + c$$

where  $a$ ,  $b$  and  $c$  are constants to be found.

(3)

The point  $M$  is the minimum turning point of  $C$ , as shown in Figure 3.

- (b) Deduce the coordinates of  $M$

(2)

The line  $l$  is the normal to  $C$  at the point  $P$ , as shown in Figure 3.

Given that  $l$  has equation  $y = k - \frac{1}{8}x$ , where  $k$  is a constant,

- (c) (i) find the coordinates of  $P$

- (ii) find the value of  $k$

(6)

Question 9 continues on the next page

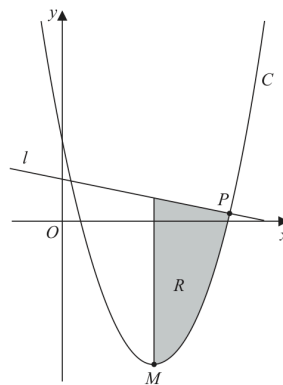


Figure 4

Figure 4 is a copy of Figure 3. The finite region  $R$ , shown shaded in Figure 4, is bounded by  $l$ ,  $C$  and the line through  $M$  parallel to the  $y$ -axis.

- (d) Identify the inequalities that define  $R$ .

(3)

(Total for Question 9 is 14 marks)

## Worked Solution - Question 9

**1. Complete the square**

$$\frac{1}{2}x^2 - 10x + 22 = \frac{1}{2}(x^2 - 20x) + 22 = \frac{1}{2}((x - 10)^2 - 100) + 22.$$

**2. Simplify the completed square form**

$$\frac{1}{2}((x - 10)^2 - 100) + 22 = \frac{1}{2}(x - 10)^2 - 28.$$

**3. Read the minimum point**

The minimum point is  $M(10, -28)$ .

**4. Use the normal gradient**

The line  $l$  has gradient  $-\frac{1}{8}$ , so the tangent gradient at  $P$  is 8.

**5. Differentiate the curve**

$$\frac{dy}{dx} = x - 10. \text{ At } P, x - 10 = 8, \text{ so } x = 18.$$

**6. Find P**

When  $x = 18$ ,  $y = \frac{1}{2}(18)^2 - 10(18) + 22 = 4$ , so  $P = (18, 4)$ .

**7. Find k**

Since  $P$  lies on  $y = k - \frac{1}{8}x$ ,  $4 = k - \frac{1}{8}(18) = k - \frac{9}{4}$ , so  $k = \frac{25}{4}$ .

**8. Write the upper boundary**

The line is  $y = \frac{25}{4} - \frac{x}{8}$ .

### 9. Write the lower boundary

The curve is  $y = \frac{1}{2}x^2 - 10x + 22$ .

### 10. Identify the region

The region lies to the right of the vertical line through  $M$  and between the curve and the normal line:  $10 \leq x \leq 18$  and  $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$ .

#### Final answer

(a)  $\frac{1}{2}(x - 10)^2 - 28$ .

(b)  $M(10, -28)$ .

(c)  $P(18, 4)$ ,  $k = \frac{25}{4}$ .

(d)  $10 \leq x \leq 18$ ,  $\frac{1}{2}x^2 - 10x + 22 \leq y \leq \frac{25}{4} - \frac{x}{8}$ .